MODEL LAB · 2026-10-11

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

29,344
held-out plate appearances

3.06%
held-out HR rate

0.5992
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-08-31, scored on the 29,344 that came after (actual HR rate 3.06%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5832	1.56×	0.02964	0.13599	0.00484
KNN comps Prior is the pooled rate of the nearest comparables.	0.5992	1.54×	0.02960	0.13502	0.00474
SVR Support-vector regression on the same feature matrix.	0.5923	1.41×	0.02964	0.13589	0.00646
Random forest Forest regression; a learned similarity metric.	0.5917	1.41×	0.02963	0.13553	0.00590
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5992	1.54×	0.02960	0.13502	0.00474
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5989	1.52×	0.02960	0.13504	0.00512
Linear regression Ordinary least squares on the same features.	0.5936	1.48×	0.02962	0.13560	0.00490
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5949	1.57×	0.02963	0.13552	0.00586
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5937	1.50×	0.02962	0.13544	0.00581
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5926	1.41×	0.02964	0.13580	0.00659

BLEND WEIGHTS (FOREST VARIANT)

KNN **0.90** SVR **0.00** RF **0.10**

SHRINKAGE K PER PRIOR

league **251** knn **251** svr **251** rf **251** core **251** ensemble **251** linear **251** logistic **251** nn **251** xgb **251**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,195 PA, actual HR rate 2.44%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5895	1.54×	0.02382	0.11470	0.01190
KNN comps Prior is the pooled rate of the nearest comparables.	0.5767	1.28×	0.02386	0.11508	0.01196
SVR Support-vector regression on the same feature matrix.	0.5606	1.15×	0.02402	0.11973	0.01671
Random forest Forest regression; a learned similarity metric.	0.5694	1.15×	0.02396	0.11638	0.01583
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5767	1.28×	0.02386	0.11508	0.01196
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5783	1.28×	0.02386	0.11509	0.01347
Linear regression Ordinary least squares on the same features.	0.5764	1.02×	0.02394	0.11808	0.01456
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5868	1.41×	0.02390	0.11586	0.01389
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5765	1.41×	0.02390	0.11596	0.01381
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5738	1.15×	0.02402	0.11848	0.01664

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Jackson Chourio 14.7%, Shohei Ohtani 14.6%, Jake Bauers 12.7%, William Contreras 12.3%, Teoscar Hernández 12.1%, Freddie Freeman 11.9%
Flat league	5/6	Jackson Chourio 14.6%, Shohei Ohtani 13.1%, William Contreras 12.1%, Jake Bauers 11.9%, Freddie Freeman 11.6%, Will Smith 11.5%
KNN comps	6/6	Jackson Chourio 14.9%, Shohei Ohtani 14.6%, Jake Bauers 12.9%, William Contreras 12.2%, Teoscar Hernández 12.2%, Freddie Freeman 11.7%
SVR	6/6	Jackson Chourio 15.3%, Shohei Ohtani 14.5%, Jake Bauers 12.9%, Teoscar Hernández 12.2%, William Contreras 12.1%, Freddie Freeman 11.6%
Random forest	5/6	Jackson Chourio 15.4%, Shohei Ohtani 14.9%, Jake Bauers 12.8%, Teoscar Hernández 12.5%, William Contreras 12.1%, Will Smith 12.1%
Core (KNN+SVR)	6/6	Jackson Chourio 14.9%, Shohei Ohtani 14.6%, Jake Bauers 12.9%, William Contreras 12.2%, Teoscar Hernández 12.2%, Freddie Freeman 11.7%
Forest blend	6/6	Jackson Chourio 14.9%, Shohei Ohtani 14.6%, Jake Bauers 12.9%, Teoscar Hernández 12.2%, William Contreras 12.2%, Freddie Freeman 11.7%
Linear regression	5/6	Jackson Chourio 15.1%, Shohei Ohtani 14.5%, Jake Bauers 12.8%, Teoscar Hernández 12.4%, Will Smith 12.2%, William Contreras 12.1%
Logistic regression	5/6	Jackson Chourio 15.2%, Shohei Ohtani 14.9%, Jake Bauers 12.7%, Teoscar Hernández 12.4%, William Contreras 12.2%, Will Smith 12.0%
Neural net (torch)	5/6	Jackson Chourio 15.2%, Shohei Ohtani 14.9%, Jake Bauers 12.8%, Teoscar Hernández 12.4%, William Contreras 12.2%, Will Smith 12.1%
XGBoost	5/6	Jackson Chourio 15.4%, Shohei Ohtani 15.1%, Jake Bauers 13.1%, Teoscar Hernández 12.4%, William Contreras 12.3%, Will Smith 11.9%

Hitter by hitter

The top 40 bats by the served probability.

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1 Jackson Chourio MIL slate	**14.7%**	14.6%	14.9%	15.3%	15.4%	14.9%	14.9%	15.1%	15.2%	
2 Shohei Ohtani LAD slate	**14.6%**	13.1%	14.6%	14.5%	14.9%	14.6%	14.6%	14.5%	14.9%	
3 Jake Bauers MIL	**12.7%**	11.9%	12.9%	12.9%	12.8%	12.9%	12.9%	12.8%	12.7%	
4 William Contreras MIL	**12.3%**	12.1%	12.2%	12.1%	12.1%	12.2%	12.2%	12.1%	12.2%	
5 Teoscar Hernández LAD	**12.1%**	11.5%	12.2%	12.2%	12.5%	12.2%	12.2%	12.4%	12.4%	
6 Freddie Freeman LAD	**11.9%**	11.6%	11.7%	11.6%	11.5%	11.7%	11.7%	11.9%	11.7%	
7 Will Smith LAD	**11.8%**	11.5%	11.6%	11.4%	12.1%	11.6%	11.7%	12.2%	12.0%	
8 Mookie Betts LAD	**10.9%**	10.9%	10.7%	11.2%	10.8%	10.7%	10.7%	11.0%	11.0%	
9 Brice Turang MIL	**10.5%**	11.0%	10.5%	10.7%	10.5%	10.5%	10.5%	10.0%	10.0%	
10 Garrett Mitchell MIL	**9.8%**	9.2%	9.5%	8.6%	9.0%	9.5%	9.5%	9.1%	9.0%	
11 Andy Pages LAD	**9.0%**	8.5%	8.8%	8.5%	8.6%	8.8%	8.7%	8.5%	8.4%	
12 Kyle Tucker LAD	**8.5%**	8.6%	8.4%	8.5%	8.5%	8.4%	8.4%	8.4%	8.2%	
13 Enrique Hernández LAD	**8.2%**	8.0%	7.9%	7.7%	7.9%	7.9%	7.9%	8.0%	7.7%	
14 Andrew Vaughn MIL	**7.9%**	9.0%	8.3%	7.3%	7.3%	8.3%	8.2%	7.2%	7.3%	
15 Joey Ortiz MIL	**5.6%**	6.6%	5.7%	5.7%	5.8%	5.7%	5.7%	5.7%	5.7%	
16 Cooper Pratt MIL	**5.4%**	7.1%	5.5%	4.8%	5.3%	5.5%	5.5%	4.8%	4.9%	
17 Luis Lara MIL	**4.2%**	6.0%	4.4%	3.3%	4.0%	4.4%	4.4%	3.5%	4.1%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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